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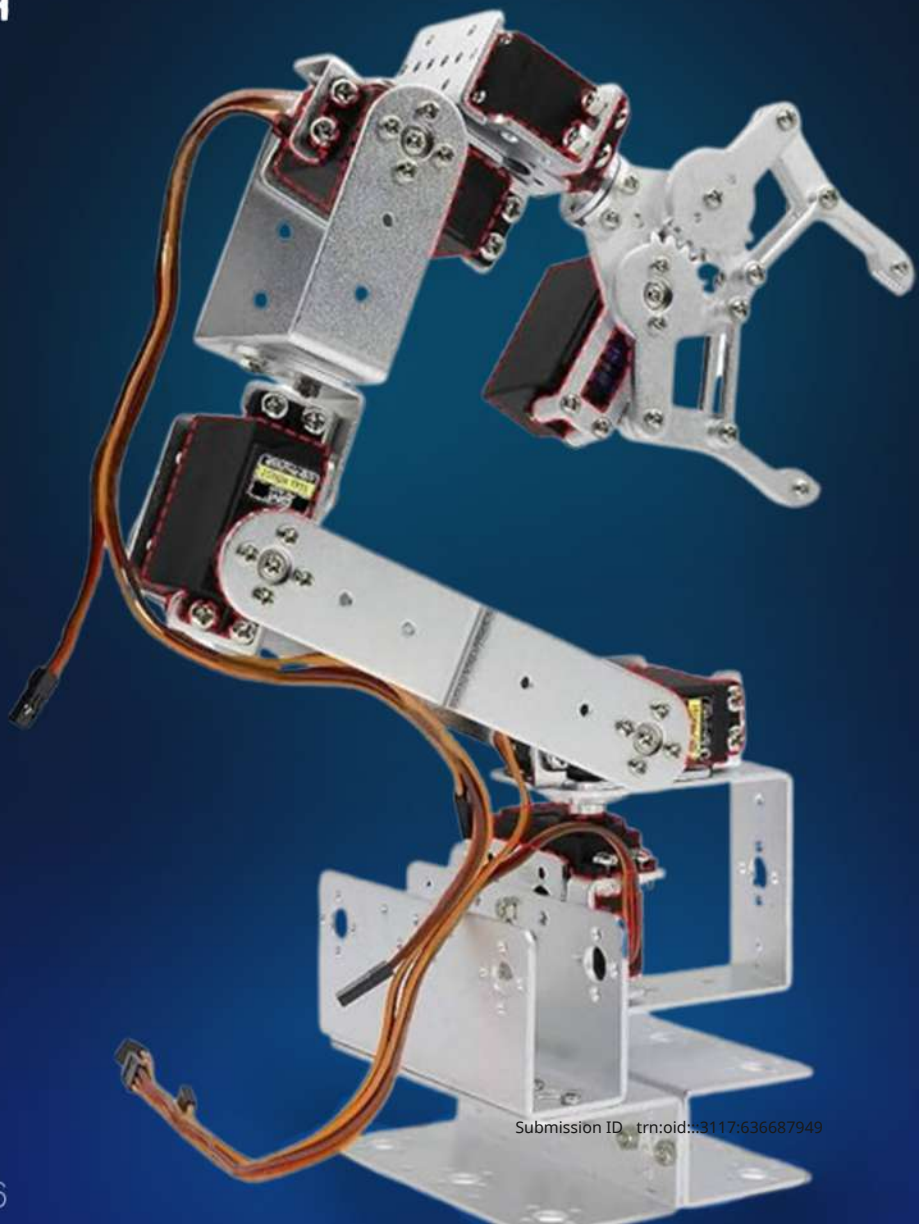
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INVESTIGATING THE FEASIBILITY OF A LOW-COST MICROCONTROLLER-BASED ASSISTIVE ROBOTIC ARMS

INTERNATIONAL ACADEMIC RESEARCH COMPETITION



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Introduction

People with limited mobility and physical impairments, including those resulting from spinal cord injury or multiple sclerosis [1], may find performing everyday activities difficult, particularly activities that require reaching, grabbing, or moving objects in general. These difficulties can affect a person's ability to independently retrieve and handle objects during everyday activities, potentially increasing the need for assistance from others.

Therefore, assistive technologies have been investigated as a way to support greater independence in these activities. Statistics also indicate that there are more than 600 million people worldwide with disabilities, and emerging technologies are beginning to adapt to the needs of people with disabilities [2].

Existing Literature

Robotic arms are one type of technology that can provide physical assistance and object manipulation [1]. Previous research has explored various ways of controlling the robot and sensing systems. For example, using an Arduino platform and a mobile application via Bluetooth, allowing the mobile user to remotely control the movement of individual servo motors [3] . Other research has investigated the use of camera and image processing techniques to identify objects and determine their positions, allowing a robotic arm to perform more controlled grasping tasks [2]. Furthermore, another research discussed how robotic arms can serve as the physical component of embodied artificial intelligence, enabling AI systems to perform physical tasks, gather information, and learn through interaction [4] .

Research Gap

Despite these developments, the cost and the complexity of some robotic system may make them difficult to implement as accessible assistive technologies. This creates an opportunity to investigate whether a low-cost robotic arm using readily available microcontrollers, components, modules and control methods can perform selected tasks relevant to assistive applications.

This research will investigate and focus on the feasibility of a low-cost 4-DOF (four-degree-of-freedom) microcontroller-based robotic arm for object-retrieval and grasping tasks. The four degrees of freedom refer to the independent movements available through the arm's joints, allowing the robotic arm to move and position its gripper within its working space [4]. The proposed system will be designed and engineered to perform controlled movements and interact with selected objects under defined testing conditions. This proposed design will consider factors such as positioning accuracy, grasping capability, movement control, and repeatability when developing the prototype. These considerations will be used to examine the feasibility and limitations of the proposed approach for potential assistive applications.

Research Questions

This particular research will be guided by the following main research question along with supporting questions to further discuss about the topic of assistive robotic arm.

Main Research Question:

How can a low-cost 4-DOF microcontroller-based robotic arm be designed and controlled to perform selected object-retrieval and grasping tasks for potential assistive applications?

Supporting Questions:

- What hardware components and sensing systems are required to develop the proposed robotic arm?
- How can the movement of the robotic arm and gripper be controlled to perform selected object-retrieval and grasping tasks?
- How can mathematical modelling and inverse kinematics be incorporated into the control of the robotic arm?
- What practical limitations should be considered when developing a low-cost 4-DOF robotic arm for potential assistive applications?

Proposed Methodology

I. Hardware Development:

The proposed robotic arm will be developed using a low-cost 4-DOF robotic arm [5] with 3 SG90 servo motors each for (a) base joints (b) shoulder joints (c) forearm joints. Then, a singular MG90 servo motor will be used for the gripper. An external 5 V power supply will be used to provide sufficient power to the servo motors, and PCA9685 16 Channel 12-bit PWM Servo Motor Driver will also be used to generate and manage the control signals for the servo motors [6]. A HC-SR04 ultrasonic sensor mounted on the gripper for the robotic arm to assist with detecting objects during grasping. A camera may also be incorporated to provide visual information about the position of selected objects. The components will then be assembled and connected using (1) Male-to-Male 10cm jumper wires (2) Male-to-Male 20cm jumper wires (3) Female-to-Male 10cm jumper wires (4) Female-to-Male 20cm jumper wires and a Type-B-data-transfer cable. These connections will allow the microcontroller to communicate with the PWM servo motor driver and control the movement of the robotic arm's joints and grippers.

II. Software Development & Control:

A. Programming Language and Software

The proposed control system will use Python for computer-side software development, including the graphical user interface, communication with the web server, AI API integration, and any planned image-processing functions. Arduino C/C++ will be used on the Arduino UNO microcontroller board to receive commands from the computer and control the PCA9685 PWM servo motor driver. The microcontroller will therefore handle the low-level movement of the robotic arm, while Python will provide the higher-level control interface.

B. Manual GUI Control

A Python-based graphical user interface will be programmed to provide direct manual control of the robotic arms. The GUI will include controls for the base, shoulders, forearm, and the gripper, allowing the user to adjust the corresponding joint positions. Commands from the GUI will be transmitted to the microcontroller board through a serial communication connection [7]. The microcontroller will then communicate with the PCA9685 PWM servo motor driver using I2C communication to generate the appropriate PWM signals for the servo motors.

C. AI API, Camera and Object Positioning

The AI-based control system will integrate a camera with an AI API to provide visual information for the robotic arm. A Full-HD Web Cam PK-920H manufactured by A4tech will be connected directly to the computer and it will capture images of the selected objects within the working area of the robotic arm. These images will then be processed and transmitted to the AI API, which will identify the relevant object and determine the positioning of the robotic arm to move the arm towards the object or activating its gripper. The command will be sent through Python-based web server to the Arduino Uno microcontroller. For more precise positioning, the object's image coordinates will be converted into coordinates relative to the robotic arm's working area before inverse kinematics is applied. The calculated joint angles will then be converted into servo commands and transmitted to the PCA9685 PWM servo motor driver.

D. Mathematical Modelling and Inverse Kinematics

Mathematical modelling will be incorporated into the control system to determine the require angles of movements of the robotic arm. Trigonometric relationships and inverse kinematics will be used to calculate suitable joint angles from the desired position of the gripper [8] [9]. The calculations will consider the lengths of the arm links, the coordinates of the object targeted, and the required position of the end effector [10]. The calculated angles will then be converted into servo commands and transmitted to the microcontroller for movement control via PCA9685 PWM servo motor driver.

1. **Calculating an elbow/forearm joint angle**

For a simplified two-link intersection of the robotic arm, inverse kinematics can be used to calculate the required joint angle from the desired position of the gripper:

$$\theta_2 = \cos^{-1}\left(\frac{x^2 + y^2 - {L_1}^2 - {L_2}^2}{2L_1L_2}\right)$$

Where:

- x, y = desired position of the gripper
- L_1 = length of the upper-arm link
- L_2 = length of the forearm link
- θ_2 = calculated angle of the second joint

This equation determines the angle required at the second joint so that the end of two-link can reach a specified position. The equation uses cosine rule, which allows the joint angle to be calculated from the link lengths and the desired gripper position.

2. **Calculating the first joint angle**

The angle of the first joint can be calculated using:

$$\theta_1 = atan2(y, x) - atan2(L2\sin(\theta_2)L1 + L2\cos(\theta_2)$$

Where:

- θ_1 = angle of the first joint
- x, y = desired gripper position
- L_1, L_2 = lengths of the two arm links
- θ_2 = angle calculated from the previous equation (given in example 1)

This equation determines the first joint angle needed to position the gripper at the selected coordinates. The "atan2" function is useful and essential because it considers the direction of both coordinates and therefore handles different quadrants more reliably than a simple inverse tangent calculation.

3. **Calculating base rotation**

For a simplified 3D (three dimensional) positioning system, the base rotation can be calculated from the horizontal coordinates:

$$\theta_{Base} = atan2(y, x)$$

Where:

- x, y = horizontal and sideways distance from the base of the robotic arm to the target position
- θ_{Base} = rotation angle of the robotic arm's base joint

15

This determines the rotational angle of the base required to orient and adjust the arm toward the target position. The calculated angle can then be converted into a servo position and sent to the motor controller.

Expected Outcomes and Potential Impacts

8

The proposed development is expected to be able to produce a functional, low-cost 4-DOF robotic-arm prototype capable of controlled movement and object grasping. The system will combine a manual GUI-based control, mathematical positioning calculations, and potentially camera-based object positioning and AI-assistant commands. This project is expected to demonstrate whether these relatively low-cost technologies can be integrated into a practical assistive robotic arm. If successfully developed, the prototype could provide a foundation for future systems, systems designed to assist individuals with limited mobility in retrieving and handling everyday objects, potentially supporting greater independence in daily activities.

Limitations

The proposed system may be limited by the arm's number of degrees of freedom, servo torque, payload capacity, working range, and mechanical accuracy. Camera-based positioning may be affected by lighting and objects' appearance, whilst ultrasonic sensing may have limitations with certain objects. The use of AI API may additionally depend on internet connectivity and the selected service. Therefore, the prototype will represent an initial feasibility-oriented design rather than a clinical assistive device.

Project work plan

The project will firstly begin with research and hardware assembly, which is expected to take approximately one month. This will be followed by microcontroller and servo-driver programming together with the development of manual GUI control and inverse kinematics calculations, which will take approximately 3 weeks. Camera-based object positioning and AI-assistant control will then be programmed and integrated over approximately 2 months. Finally, the completed prototype will undergo functional checks, and its design, limitations, and feasibility will be documented.

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